Rockchip RK3506 Linux6.1 SDK ReleaseNote

ID: RK-FB-YF-A25

Release Version: V1.1.0

Release Date: 2024-11-28

Security Level: □Top-Secret □Secret □Internal ■Public

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Preface

Overview

The document presents Rockchip RK3506 Linux6.1 SDK release note, aiming to help engineers get started with RK3506 Linux6.1 SDK development and debugging faster.

Intended Audience

This document (this guide) is mainly intended for:

Technical support engineers

Software development engineers

Chipset and System Support

Chipset	Buildroot	Debian	Yocto
RK3506	Y	N	Y

Revision History

Date	Version	Author	Revision History
2024-07-20	V0.0.1	Zack.Huang	Alpha version
2024-08-20	V0.1.0	Zack.Huang	Beta version
2024-10-20	V1.0.0	Zack.Huang	Release version
2024-11-28	V1.1.0	Zack.Huang	Release v1.1.0 version

Contents

1. Overview

This SDK supports systems based on Buildroot 2024.02, with the kernel based on Kernel 6.1 and the bootloader based on U-boot v2017.09. It is suitable for RK3506 EVB development boards and all Linux products developed based on this board.

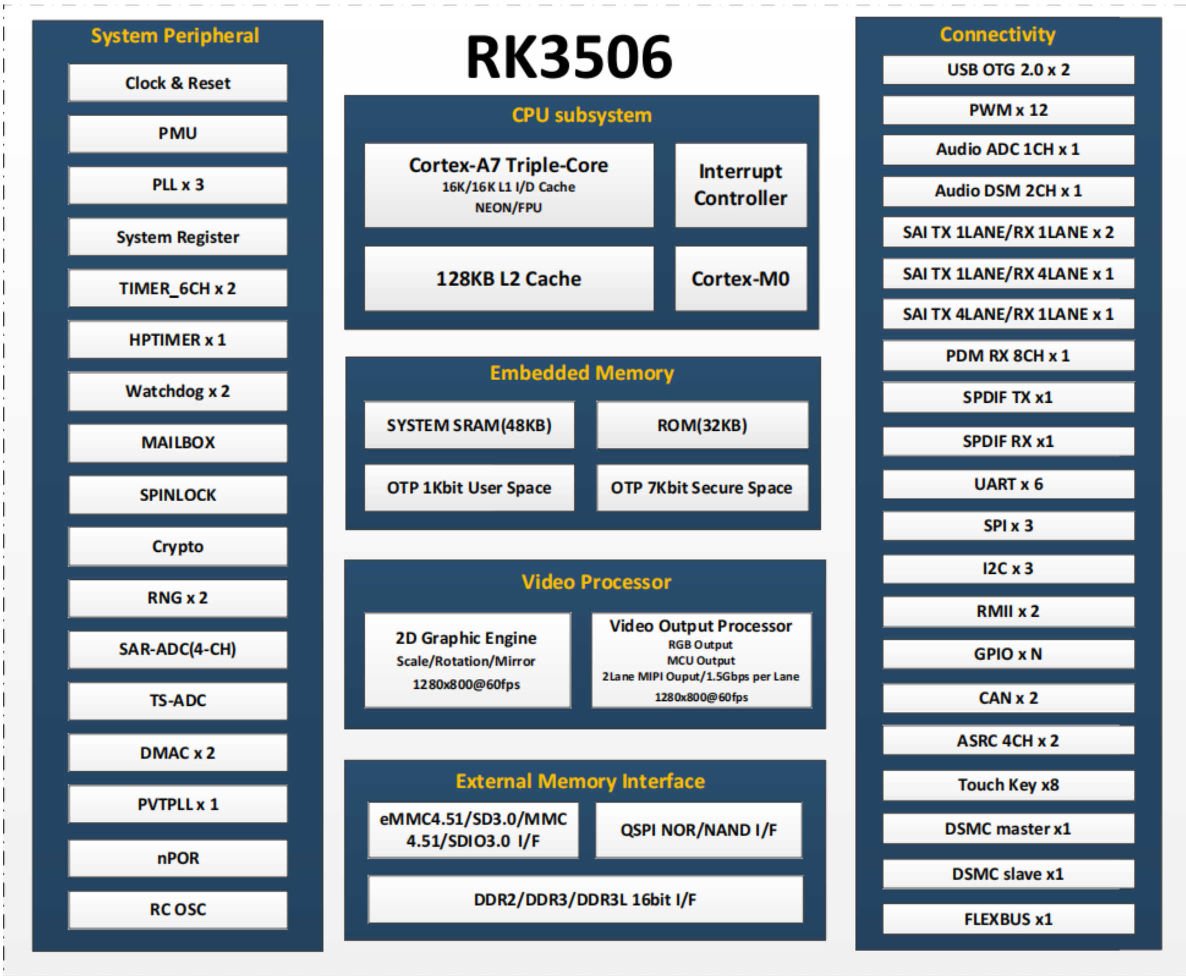
The SDK is suitable for AIoT products such as HMI, apartment video intercom, PLC, etc. providing flexible data path combination interfaces to meet the customized requirements for free combination, please refer to the documents under the project's docs/ directory.

2. Main Functions

Function	Module Name
System	Buildroot,Amp(RT_Thread + Linux + HAL),Yocto
Partition table	uboot, misc, boot, recovery, rootfs, oem, userdata
File System Type	EXT2/3/4, VFAT, NTFS, UBIFS, SquashFS
Upgrade Recovery	OTA, AB,Recovery
Secure Boot	SecureBoot
Stress Test Tool	ROCKCHIP_TEST
Data communication	Wi-Fi, Ethernet card, USB, SD card
Applications	lvgl demo for home appliance display and control

2.1 Hardware functions

For specific hardware interface functions, please refer to the RK3506 chip diagram below



3. How to Get the SDK

The SDK is released by Rockchip server. Please refer to Chapter 4 [Software Development Guide] to build a development environment.

3.1 Get Source Code from Rockchip Code Server

To get RK3506 Linux SDK software package, customers need an account to access the source code repository provided by Rockchip. In order to be able to obtain code synchronization, please provide SSH public key for server authentication and authorization when apply for SDK from Rockchip technical window. About Rockchip server SSH public key authorization, please refer to Chapter 6 SSH Public Key Operation Introduction.

RK3506 Linux SDK download command is as follows：

```
repo init --repo-url https://gerrit.rock-chips.com:8443/repo-release/tools/repo \
-u https://gerrit.rock-chips.com:8443/linux/rockchip/platform/manifests -b \
rk3506 -m rk3506_linux6.1_release.xml
```

Repo, a tool built on Python script by Google to help manage git repositories, is mainly used to download and manage software repository of projects. The download address is as follows:

```
git clone https://gerrit.rock-chips.com:8443/repo-release/tools/repo
```

3.2 Get Source Code from Local Compression Package

For quick access to SDK source code, Rockchip Technical Window usually provides corresponding version of SDK initial compression package. In this way, developers can get SDK source code through decompressing the initial compression package, which is the same as the one downloaded by repo.
Take RK3506_LINUX6.1_SDK_Release_V1.0.0_20241020.tgz as an example. After geting a initialization package, you can get source code by running the following command:

```
mkdir rk3506
tar xvf RK3506_LINUX6.1_SDK_Release_V1.0.0_20241020.tgz -C rk3506
cd rk3506
.repo/repo/repo sync -l
.repo/repo/repo sync -c
```

Developers can update via `.repo/repo/repo sync -c` command according to update introductions that are regularly released by FAE window.

If you encounter issues downloading the repository, you can use the `--force-sync` parameter to force an update, as in `.repo/repo/repo sync -c --force-sync`. Before doing so, please ensure that any local modifications have been backed up.

After updating the SDK code, a clean operation is required. For example: `./build.sh cleanall`

> Note:
> The software release version can be viewed through engineering XML, as follows:

```
.repo/manifests$ realpath rk3506_linux_release.xml
e.g., the printed version is v1.0.0 and the update time is 20241020
<SDK>/.repo/manifests/release/rk3506_linux6.1_release_v1.0.0_20241020.xml
```

4. Software Development Guide

For software development, please refer to the quick start documents in the project directory:

```
<SDK>/docs/en/RK3506/Quick-start/Rockchip_RK3506_Quick_Start_Linux_EN.pdf
```

5. Hardware Development Guide

For hardware development, please refer to the following documents in the project directory:

```
<SDK>/docs/en/Socs/RK3506/Hardware/
```

6. SSH Public Key Operation Introduction

Please follow the introduction in the “Rockchip_User_Guide_SDK_Application_And_Synchronization_EN” to generate an SSH public key and send the email to sw.fae@rock-chips.com, to get the SDK code.
This document will be released to customers during the process of applying for permission.

6.1 Key Authority Management

Server can monitor download times and IP information of a key in real time. If an abnormality is found, download permission of the corresponding key will be disabled.

Keep the private key file properly. Do not grant second authorization to third parties.

6.2 Reference Documents

For more details, please refer to document
“/docs/en/Others/Rockchip_User_Guide_SDK_Application_And_Synchronization_EN.pdf"